Linear Regression

- using the boston housing data, do a multiple linear regression of the house price
  1. How good is the fit (R^2)?
  2. Are there variables for which null hypothesis is valid? (with confidence interval containing 0 or p > 0.05).
  3. Now do a multiple linear regression using all the variables except INDUS and AGE which were found to be not influential. How does it's R^2 compare with the case when using all variables.
  4. Is there a variable for which null hypothesis is valid? (with confidence interval containing 0 or p > 0.05).
  5. If we further take out the distance DIST, is the fit badly affected?

In [9]:

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')

# Load Boston Housing Dataset
data = pd.read_csv(r"C:\Users\31064\Downloads\archive (1)\boston.csv")

# Separate features and target
X = data.drop('MEDV', axis=1)
y = data['MEDV']

# QUESTION 1: Multiple Linear Regression with ALL variables
X_model1 = sm.add_constant(X)
model1 = sm.OLS(y, X_model1).fit()

r2_model1 = model1.rsquared
adj_r2_model1 = model1.rsquared_adj

print(f"R² = {r2_model1:.4f}")
print(f"Adjusted R² = {adj_r2_model1:.4f}")
```

```
R² = 0.7406
Adjusted R² = 0.7338
```

In [12]:
```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')

# Load Boston Housing Dataset
data = pd.read_csv(r"C:\Users\31064\Downloads\archive (1)\boston.csv")

# Separate features and target
X = data.drop('MEDV', axis=1)
y = data['MEDV']

# QUESTION 2: Are there variables for which null hypothesis is valid?
#             (p > 0.05 or confidence interval containing 0)

X_model1 = sm.add_constant(X)
model1 = sm.OLS(y, X_model1).fit()

print("Variables where null hypothesis is valid (p > 0.05 or CI contains 0):")
print("="*70)

non_sig_vars = []

for var in X.columns:
    p_value = model1.pvalues[var]
    ci_lower, ci_upper = model1.conf_int().loc[var]
    coef = model1.params[var]

    contains_zero = (ci_lower <= 0 <= ci_upper)

    if p_value > 0.05 or contains_zero:
        non_sig_vars.append(var)
        print(f"\n{var}:")
        print(f"  Coefficient: {coef:.6f}")
        print(f"  p-value: {p_value:.6f}")
        print(f"  95% CI: [{ci_lower:.6f}, {ci_upper:.6f}]")
        if contains_zero:
            print(f"  → CI contains 0")
        if p_value > 0.05:
            print(f"  → p > 0.05 (not significant)")

print(f"\n{'='*70}")
print(f"Total non-significant variables: {len(non_sig_vars)}")
print(f"Variables: {non_sig_vars}")
```

```
Variables where null hypothesis is valid (p > 0.05 or CI contains 0):
======================================================================

INDUS:
  Coefficient: 0.020559
  p-value: 0.738288
  95% CI: [-0.100268, 0.141385]
  → CI contains 0
  → p > 0.05 (not significant)

AGE:
  Coefficient: 0.000692
  p-value: 0.958229
  95% CI: [-0.025262, 0.026647]
  → CI contains 0
  → p > 0.05 (not significant)

======================================================================
Total non-significant variables: 2
Variables: ['INDUS', 'AGE']
```

In [13]:

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')

# Load Boston Housing Dataset
data = pd.read_csv(r"C:\Users\31064\Downloads\archive (1)\boston.csv")

# Separate features and target
X = data.drop('MEDV', axis=1)
y = data['MEDV']

# QUESTION 3: Regression without INDUS and AGE
#             How does R² compare with using all variables?

# Model 1: All variables
X_model1 = sm.add_constant(X)
model1 = sm.OLS(y, X_model1).fit()
r2_model1 = model1.rsquared
adj_r2_model1 = model1.rsquared_adj

# Model 2: Without INDUS and AGE
X_model2 = X.drop(columns=['INDUS', 'AGE'])
X_model2 = sm.add_constant(X_model2)
model2 = sm.OLS(y, X_model2).fit()
r2_model2 = model2.rsquared
adj_r2_model2 = model2.rsquared_adj

print("COMPARISON: Model 1 (All variables) vs Model 2 (Without INDUS, AGE)")
print("="*70)
print(f"\nModel 1 (All variables):")
print(f"  R² = {r2_model1:.6f}")
print(f"  Adjusted R² = {adj_r2_model1:.6f}")

print(f"\nModel 2 (Without INDUS, AGE):")
print(f"  R² = {r2_model2:.6f}")
print(f"  Adjusted R² = {adj_r2_model2:.6f}")

print(f"\nChange:")
print(f"  ΔR² = {r2_model2 - r2_model1:.6f}")
print(f"  ΔAdjusted R² = {adj_r2_model2 - adj_r2_model1:.6f}")
```

```
COMPARISON: Model 1 (All variables) vs Model 2 (Without INDUS, AGE)
======================================================================

Model 1 (All variables):
  R² = 0.740643
  Adjusted R² = 0.733790

Model 2 (Without INDUS, AGE):
  R² = 0.740582
  Adjusted R² = 0.734806

Change:
  ΔR² = -0.000060
  ΔAdjusted R² = 0.001016
```

In [14]:
```
import numpy as np
import pandas as pd
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')

# Load Boston Housing Dataset
data = pd.read_csv(r"C:\Users\31064\Downloads\archive (1)\boston.csv")

# Separate features and target
X = data.drop('MEDV', axis=1)
y = data['MEDV']

# QUESTION 4: In Model 2 (without INDUS and AGE),
#             is there a variable for which null hypothesis is valid?
#             (p > 0.05 or confidence interval containing 0)

# Model 2: Without INDUS and AGE
X_model2 = X.drop(columns=['INDUS', 'AGE'])
X_model2 = sm.add_constant(X_model2)
model2 = sm.OLS(y, X_model2).fit()

print("Variables where null hypothesis is valid in Model 2:")
print("(p > 0.05 or CI contains 0)")
print("="*70)

non_sig_vars = []

for var in X_model2.columns:
    if var == 'const':
        continue

    p_value = model2.pvalues[var]
    ci_lower, ci_upper = model2.conf_int().loc[var]
    coef = model2.params[var]

    contains_zero = (ci_lower <= 0 <= ci_upper)

    if p_value > 0.05 or contains_zero:
        non_sig_vars.append(var)
        print(f"\n{var}:")
        print(f"  Coefficient: {coef:.6f}")
        print(f"  p-value: {p_value:.6f}")
        print(f"  95% CI: [{ci_lower:.6f}, {ci_upper:.6f}]")
        if contains_zero:
            print(f"  → CI contains 0")
        if p_value > 0.05:
            print(f"  → p > 0.05 (not significant)")

if len(non_sig_vars) == 0:
    print("\nNO - All variables in Model 2 are statistically significant (p < 0.05)")
else:
    print(f"\n{'='*70}")
    print(f"Total non-significant variables: {len(non_sig_vars)}")
    print(f"Variables: {non_sig_vars}")
```

```
Variables where null hypothesis is valid in Model 2:
(p > 0.05 or CI contains 0)
======================================================================

NO - All variables in Model 2 are statistically significant (p < 0.05)
```

In [ ]: